Ram Leakage Geometry and Spectral-Band Transfer for the Kadison–Kaplansky Construction

Restricted Operator Improvements and Trace-Sensitive Boundaries

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Abstract

We study the operator-theoretic transfer steps in the pinned manuscript *A torsion-free counterexample to the Kadison–Kaplansky projection conjecture*. The target is a scalar projection in the reduced group C^*-algebra of a finitely generated torsion-free group, not merely a matrix projection or a projection in a von Neumann completion. Starting from the actual Ram finite-frame leakage identity and physical Gram representation, we retain the exact corner metric EJE when a reducing upper spectral space is compressed. This gives a quadratic diagonal error, a sharper full block-norm estimate, and a Schur-complement criterion preserving an upper spectral gap under quadratic leakage. We prove the corresponding equivariance-sensitive projection equivalence, give conditional cover and copy applications, and derive a norm-convergent scalar-corner inverse with explicit truncation costs. Balancing the diagonal traces extends the source's sufficient scalarization range from trace below one half to every trace strictly below one, at the cost of a larger finite free factor. Trace identities are kept separate from nonequivariant spectral comparisons. The tower, nilpotent sampling, and geometric construction are inherited inputs, not a new certification of the claimed global counterexample. Contribution assessment: C3/5, proved leakage-sensitive spectral transfer and balanced finite-free-factor scalarization strengthen specific source steps while the global group construction and regular-norm realization remain inherited.

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1 Problem, Algebra, and Quantifiers

For a discrete group Γ, let

$$(\lambda(g)\xi)(h) = \xi(g^{-1}h), \qquad C_r^*(\Gamma) = \overline{\lambda(\mathbb{C}[\Gamma])}^{\|\cdot\|} \subset B(\ell^2\Gamma).$$

The scalar projection conjecture asserts that a torsion-free Γ has only $0, 1$ as projections in this algebra. The manuscript claims a finitely generated torsion-free Γ and $e = e^* = e^2$ with $0 < \tau_\Gamma(e) < 1/2$, where

$$\tau_\Gamma(x) = \langle \delta_1, x\delta_1 \rangle, \qquad \tau_{\Gamma,N}(X) = \sum_{j=1}^{N} \tau_\Gamma(X_{jj}).$$

The matrix trace is *not divided by* N. Our inner product is conjugate-linear in its first argument. All local operator theorems below concern bounded operators on complex Hilbert spaces and use the actual operator norm, unless another norm is named.

The source is pinned at repository commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`, with manuscript date September 23, 2026 [1]. Its matrix target is a finite $T = T^* \in M_N(\mathbb{C}[\Lambda])$ in the regular representation whose nonempty separated upper band has projection

$$p_0 \in M_N(C_r^*(\Lambda)), \qquad 0 < \tau_{\Lambda,N}(p_0) < 1/2. \tag{1}$$

Scalar compression then takes place over $\Lambda * \mathbb{F}_N$, not automatically over Λ. Existence of a matrix corner alone does not achieve the scalar target. Nor does a Borel spectral projection in $L(\Lambda)$ establish membership in $C_r^*(\Lambda)$.

Faithfulness of the canonical trace is elementary but important. For $x \geq 0$ with $\tau(x) = 0$, $x^{1/2}\delta_1 = 0$. Commutation with right translations makes $x^{1/2}$ annihilate every group basis vector; hence $x = 0$. A scalar projection of trace strictly between zero and one is therefore nontrivial. Normalized matrix trace cannot replace the stated criterion: $\mathrm{diag}(1, 0, \ldots, 0)$ has normalized trace $1/N$ even when the scalar algebra has no nontrivial projections.

2 Source Chain and Its Analytic Interfaces

The manuscript makes one sufficiently large *finite* parameter choice, not a limit-group argument. Its parameters include

$$d = 2^{4\ell}, \quad D = d^2, \quad s = 10\ell, \quad r = 7\ell, \quad H_i = (2 \cdot 1000 + 1)s + 1 + |i|.$$

All $o(1)$ estimates used in the final selection must be uniform in the infinitely many layers $i \in \mathbb{Z}$. A separate ℓ for every layer would not prove the construction.

The analytic chain has the following distinct interfaces.

1. The expanding tower Y^0 and path-recoverable tower $Y^{\#}$ share a pruned induced graph Y. The deletion fraction β_ℓ is uniform.

2. A nilpotent voltage cover P realizes a central direct-integral model $O(t)$. Its upper space $E_*(t)$ is supported on at most two adjacent layers and is nonzero on $0 < t < \delta$.

3. The sampling theorem supplies a self-adjoint adjacency T_P with $\|T_P - \Pi O \Pi\| \le \eta$ in the full regular norm, while the same voltages satisfy the geometric girth requirement.

4. Compression and contour perturbation transfer the band to T_P. The equivariant polar maps preserve its fiber identity trace exactly.

5. Embedded indexed copies Q in the Cayley graph have disjoint assigned domains D_Q after finite trimming. These assignments need not be equivariant. Their remaining edges form a forest, yielding a small operator-norm error.

6. A separate equivariant restriction map using *all* indexed copies controls the canonical trace of the global upper space. Finally the free-product row produces a scalar projection from (1).

No local theorem in this report supplies the missing geometric or sampling premise merely from a numerical compression. We retain the source's parameter order and the finite matrix amplification in every trace comparison.

2.1 The transverse projection is not a finite magnetic projection

The actual proofs in `nilpotent.tex` first realize twisted Schwartz convolution on the affine nilpotent quotient $\bar{E}$ as Schwartz integral kernels on $L^2(\mathfrak{n})$. With the source's positive-sign Fourier transform, the correspondence is

$$\mathcal{K}_J(x, y) = \widehat{J}_+(\epsilon x, (1 + x)(1 + y)^{-1}).$$

The coordinate change and its inverse are polynomial with unit absolute Jacobian. Partial Fourier transform is an L^2 isometry; consequently the coefficient of a normalized rank-one Gaussian kernel is a twisted Schwartz idempotent J_ϵ of L^2 norm one. This is a continuous model, not yet a lattice projection.

For $y = |\theta|^{1/m}$ and $\nu = |\theta|^{n_{\mathrm{f}}}$, the source samples coefficients $\nu J_\epsilon(\Delta_y \gamma)$ to obtain a self-adjoint K_θ in the actual twisted reduced algebra. Cellwise mean-value bounds for weighted rectangular Riemann sums give

$$\|K_\theta^2 - K_\theta\|_{\mathrm{coeff},1} = O(y), \quad \|K_\theta^2 - K_\theta\|_{\mathrm{coeff},2} = O(y\sqrt{\nu}), \quad \|K_\theta \delta_1\| \asymp \sqrt{\nu}.$$

These norms serve different purposes. The first separates the spectrum into neighborhoods of zero and one, enabling continuous spectral rounding in the reduced algebra. If the rounded projection vanished, $\|K_\theta\| \le 1/4$ would imply

$$\|(K_\theta^2 - K_\theta)\delta_1\| \ge (3/4)\|K_\theta \delta_1\|,$$

contradicting the relative ℓ^2 defect for small y. Finally $0 \le p_\theta \le 4K_\theta^2$ gives $0 < \tau_\theta(p_\theta) \le C|\theta|^{n_{\mathrm{f}}}$. Small absolute operator defect alone would not prove nonzeroness.

The finite magnetic representation has dimension $N = q^{n_{\mathrm{f}}}$ and central character $-\delta$, where $\delta = q^{-1}$. A diagonal unitary on the quotient coordinates conjugates $u_t \otimes R$ to N copies of $u_{t-\delta}$ and fixes the identity fiber. Thus it contributes exactly the *unnormalized* multiplicity N to the two-factor projection:

$$s_p(t) = N\tau_t(p_t)\tau_{t-\delta}(p_{t-\delta}) \le CNt^{n_{\mathrm{f}}}(\delta - t)^{n_{\mathrm{f}}}, \quad 0 < t < \delta.$$

For windows $h_i(t) = h(\kappa^{-1}\log_2(t/(\delta - t)) - i)$ with $\sum_i h_i^2 = 1$, the active set contains at most two adjacent indices. The source's weighted trace calculation is explicitly

$$2^{C_\bullet H_i} s_p(t) \le Cd^{-(A_* n_{\mathrm{f}} - C_\bullet h_*)} 2^{-(\kappa n_{\mathrm{f}} - C_\bullet)|i|}, \qquad h_* = 5003.$$

Both exponents are positive by its ordered parameter choices. This is the uniform layer estimate needed later; it is not convergence on each fixed layer followed by an unjustified supremum.

2.2 How the source bridges finite tests and the regular norm

On constant vertex vectors the model's off-diagonal block is $|h(t)\rangle\langle h(t)| \otimes p(t)$; its symmetric upper copy has eigenvalue one, whereas the antisymmetric copy is at minus one. Expansion bounds the mean-zero vertex part by $1 - \chi$. Summing vertex diagonals cancels the constant-vector normalizations, so the upper trace is $s_p(t)$, not the number of vertices times $s_p(t)$. Removing a fraction at most β_ℓ in each layer gives squared leakage at most β_ℓ.

The full realization theorem in `sampling.tex` is stronger than testing selected finite-dimensional representations. Its proof uses graded nilpotent coordinate boxes $\mathcal{B}_x$ with uniform boundary fraction $O(x^{-B})$ for every possible translation support. For a self-adjoint convolution matrix with coefficient row bound R, averaging over left translates of the box multiplies its coefficient at g by $\alpha_x(g) = |\mathcal{B}_x \cap \mathcal{B}_x g^{-1}|/|\mathcal{B}_x|$. Every compressed quadratic form has the same norm bound; the Schur estimate for the omitted coefficients then gives

$$\|P_x A P_x\| \le \|A\| \le \|P_x A P_x\| + Cx^{-B}R.$$

The boundary term is an analytic premise, not an empirical extrapolation.

Compact coefficient templates are rounded only in their coefficients: the twisting unitaries retain the actual t. A mask of width $w_i = q^{-1}2^{-\kappa|i|}$ has Fourier coefficients bounded by $C_j w_i(1 + |n|w_i)^{-j}$. Truncating at L/w_i therefore has a uniform tail $O((1+L)^{1-j})$ on the entire circle, including small but nonzero leakage outside the window. Congruence-subgroup factorization supplies one common matrix list for the far layers; the resulting finite weight alphabet is $O_\eta(d^{89/32}(1 + \log d)^3) = o(dD)$.

With one independent voltage draw per edge, centering kills every moment word with a singleton edge. The source's rooted-core encoding includes cycle edges and yields $C_b^l \Delta^l$ remaining words at bounded cycle rank. The finite test dimension is $2^{C_1 H_i}$, and with $l = \lceil C_2 H_i / \log_2 d \rceil$ the block failure probability is bounded by $2^{C_1 H_i}(C_5/d)^l \le 2^{-4H_i}$ after a single large choice of ℓ. The same draws satisfy the girth tests with summable failure probabilities via nonzero nilpotent word polynomials. These two summable budgets permit a simultaneous choice over all layers. This report has read that proof, but the local refinements below do not independently certify its entire probabilistic or geometric chain.

2.3 The geometric and torsion boundary

The proof in `geometry.tex` requires source-path assignments through reduced diagrams, not just matching boundary labels. Short intersection arcs and the face-count estimate establish embedded copies and the tree-like trimmed intersections. Its all-copy incidence identification retains possible copy stabilizers, which inject into the torsion-free nilpotent deck group. The coned complex is acyclic over every field. A hypothetical finite cyclic prime-order subgroup would act freely on its cells and give a length-two free resolution, contradicting the nonzero higher homology of that cyclic group over its prime field. These are inherited geometric obligations; neither a finite positive matrix nor our corner formulas imply them.

2.4 The particular local inequality to improve

The source compression lemma assumes

$$A = A^*, \quad \|A\| \le 2, \quad AE = EA, \quad \|(A-I)E\| \le a, \quad A|_{(I-E)\mathcal{H}} \le (1-b)I, \quad 0 < b < 1. \tag{2}$$

For an orthogonal deletion projection J it sets $\varepsilon = \|(I-J)E\| \le 1/2$, forms the range projection F of JE, and compares JAJ on $J\mathcal{H}$ with $B_J = F + (J-F)A(J-F)$. Its sufficient estimate is $\|JAJ - B_J\| \le a + 20\varepsilon$. The proof actually gives $a + 12\varepsilon$ before relaxing the constant. The next sections preserve this exact setup and improve the estimate; they do not replace the whole source construction.

3 The Actual Ram Leakage and Physical Metric

The private TR248 proof [2] starts with measurable windows w_p satisfying $\sum_p w_p^2 = 1$ and complete local orthonormal families. For local retention projections Π_p, multiplication operators M_p, and retained analysis $\mathcal{F}$, it proves

$$\mathcal{F}^*\mathcal{F} = I - L, \qquad L = \sum_p M_p(I - \Pi_p)M_p \ge 0. \tag{3}$$

Indeed local Parseval gives the complete-bank norm identity, and replacing each identity by Π_p gives (3). For an isometric synthesis U onto a specified model, $T = \mathcal{F}U$ has $T^*T = I - U^*LU$. If $\|U^*LU\| < 1$, its inverse on the model has norm at most $(1 - \|U^*LU\|)^{-1/2}$. Off-model aliasing remains: writing $f = Ux + r$, observed data $c = \mathcal{F}f + e$, and $D = (T^*T)^{-1}T^*$ gives

$$\|UDc - f\| \le \|r\| + \frac{\|\mathcal{F}r\| + \|e\|}{\sqrt{1 - \|U^*LU\|}}.$$

This follows by subtracting f and applying the triangle inequality.

For the present source the exact analogue is restriction of J to $E\mathcal{H}$, whose Gram operator is $G = EJE$ and whose leakage is $L_E = E(I-J)E$. This is an operator on the full upper space, possibly infinite dimensional; no finite list of Ram atoms is presumed to measure that space injectively. The borrowed content is the retained physical metric and leakage account, not an invented transform diagonalization.

Volume 21 [3, 4] makes the same distinction for an injective finite synthesis $S : \mathbb{C}^n \to \mathcal{H}$: $G_S = S^*S$ is the physical Gram matrix, $P_S = SG_S^{-1}S^*$, and $B = G_S^{-1}S^*AS$. Direct orthogonal decomposition yields

$$\|(A-z)Sc\|^2 = \|(B-z)c\|_{G_S}^2 + \|(I-P_S)ASc\|^2 . \tag{4}$$

The second term cannot be deleted because a finite eigenvalue agrees with a desired band. For unbounded operators the columns must also lie in the actual domain. Here all source adjacency operators are bounded.

4 Exact Leakage Geometry and a Sharper Compression

Theorem 4.1 (Leakage-sensitive block bound)**.** *Assume* (2) *with* $a \geq 0$. *Let* $J = J^* = J^2$ *and* $\varepsilon = \|(I - J)E\| < 1$. *On* $E\mathcal{H}$ *set*

$$G = EJE, \qquad V = JEG^{-1/2}, \qquad F = VV^*, \quad Q = I - E, \quad R = J - F, \quad W = QV.$$

All inverses are in the E *corner. Then* $V^*V = E$, $F \leq J$, *and, with* E *the identity on* $E\mathcal{H}$,

$$EV = G^{1/2}, \quad W^*W = E - G, \quad \|W\| = \varepsilon, \quad RE = RV = RW = 0. \tag{5}$$

Under the unitary identification $E\mathcal{H} \oplus R\mathcal{H} \to J\mathcal{H}$, $(x, y) \mapsto Vx + y$, *write*

$$JAJ = \begin{pmatrix} K & C^* \\ C & D \end{pmatrix}, \qquad B_J = \begin{pmatrix} E & 0 \\ 0 & D \end{pmatrix}.$$

Then, putting $d_0 = a + 3\varepsilon^2$ *and* $c_0 = (3 - b)\varepsilon/2$,

$$(1 - d_0)E \leq K \leq (1 + a)E, \qquad \|K - E\| \leq d_0, \qquad\qquad D \leq (1 - b)R, \tag{6}$$

$$\|C\| \leq c_0, \qquad \|JAJ - B_J\| \leq h(d_0, c_0) := \frac{d_0 + \sqrt{d_0^2 + 4c_0^2}}{2}. \tag{7}$$

The construction commutes with every unitary or projective unitary action that commutes with A, E, J. *It lies in their generated unital* C^**-algebra, with corner operations interpreted there.*

Proof. $E - G = E(I - J)E = ((I - J)E)^*((I - J)E)$, so $(1 - \varepsilon^2)E \leq G \leq E$. The inverse square root exists in the corner, and $V^*V = G^{-1/2}GG^{-1/2} = E$. Thus F projects onto the closed range $JE\mathcal{H}$. Since $EV = GG^{-1/2} = G^{1/2}$, $W^*W = V^*(I - E)V = E - G$, proving the exact norm assertion. For $y \in R\mathcal{H}$, $Jy = y$ and $y \perp JE\mathcal{H}$, hence $\langle y, Ex\rangle = \langle y, JEx\rangle = 0$. Thus $RE = 0$. Also $RV = 0$, whence $RW = R(V - EV) = 0$.

Let $A_E = A|_{E\mathcal{H}}$ and $A_Q = A|_{Q\mathcal{H}}$. Reduction by E gives

$$K - E = G^{1/2}(A_E - E)G^{1/2} + W^*(A_Q - I)W. \tag{8}$$

The first summand lies between $-aG$ and aG. The second is between $-3(E - G)$ and $-b(E - G)$, because $-2I \leq A_Q \leq (1 - b)I$. This proves both order bounds and $\|K - E\| \leq a + 3\varepsilon^2$. The estimate is quadratic on the diagonal because E reduces A; without that premise cross terms need not vanish.

The off-diagonal block is $C = RAV = RA_QW$. Set $m = (-1 - b)/2$. Since $RW = 0$,

$$C = R(A_Q - mI)W, \qquad \|A_Q - mI\| \leq (3 - b)/2.$$

This proves $\|C\| \leq c_0$. For any x, y, the absolute quadratic form of the difference block is at most $d_0\|x\|^2 + 2c_0\|x\|\,\|y\|$. The largest eigenvalue of $\left(\begin{smallmatrix} d_0 & c_0 \\ c_0 & 0 \end{smallmatrix}\right)$ is $h(d_0, c_0)$. Taking the supremum over unit vectors proves (7). Finally $D = RAR$ inherits the complementary upper bound since $R\mathcal{H} \subset Q\mathcal{H}$. Functional calculus and all products used preserve the specified commutants. The inverse square root can be computed as the restriction of the continuous function of $G + (I - E)$, so no membership in an unspecified larger algebra is used. □

If $E = 0$, all its corner maps are the unique zero maps, $F = 0$, and $JAJ = B_J$; the zero-corner case needs no nonzero inverse. The theorem does not assert a nonzero upper band unless $E \neq 0$.

The source uses $\varepsilon \leq 1/2$, whereas the exact geometric formulas only require $\varepsilon < 1$. With $a = 0$ and fixed b, the leading coefficient in (7) is $(3-b)/2$, not 20. More importantly, its diagonal interval has width $O(\varepsilon^2)$. If the stronger source-model bound $\|A\| \leq 1$ is available, exactly the same proof replaces 3 by 2 and c_0 by $(2-b)\varepsilon/2$. These substitutions are confined to that stronger hypothesis, not applied silently to an actual cover operator known only to have norm at most two.

5 A Quadratic Gap Criterion

The norm bound is convenient for reusing the source's contour lemma. It is not the only way to preserve the band: an off-diagonal block repels the spectra of ordered diagonal blocks. The following proof is a direct Schur-complement argument, in the classical setting studied by Davis–Kahan and later block-operator work [6].

Theorem 5.1 (Gap, upper-band localization, and equivalence)**.** *In Theorem 4.1, suppose*

$$d_0 = a + 3\varepsilon^2 < b. \tag{9}$$

Then JAJ on $J\mathcal{H}$ has no spectrum in $(1-b, 1-d_0)$. Its projection P_+ above any number in that interval is equivalent to F by a partial isometry in the unital C^*-algebra generated by the blocks. The equivalence preserves every common equivariance. In particular $P_+ \neq 0$ if $E \neq 0$. Define

$$s_+ = \frac{\sqrt{(b+a)^2 + 4c_0^2} - (b+a)}{2}.$$

The upper spectral part lies in

$$[1 - d_0,\ 1 + a + s_+], \qquad s_+ \leq \frac{c_0^2}{b+a}. \tag{10}$$

If A' is self-adjoint on $J\mathcal{H}$, $\|A' - JAJ\| \leq u$, and $2u < b - d_0$, the corresponding upper band persists, nonempty if $E \neq 0$, with a gap $(1-b+u, 1-d_0-u)$ and upper interval $[1-d_0-u, 1+a+s_++u]$. Its projection remains equivalent to F.

Proof. For real $z \in (1-b, 1-d_0)$, the block $D - z$ is negative and invertible. The Schur complement is

$$K - z - C^*(D-z)^{-1}C \ \geq\ K - z \ \geq (1 - d_0 - z)E > 0.$$

It is boundedly invertible; the usual two triangular block factors give a bounded inverse for $JAJ - z$. This argument remains valid after replacing C by tC for every $0 \leq t \leq 1$. Fix the midpoint $z_* = (2 - b - d_0)/2$. The block paths are norm-continuous, uniformly bounded, and avoid a common open interval about z_*. A single continuous function equal to zero below that interval and one above it defines a norm-continuous projection path P_t in the same algebra. At $t = 0$, $P_0 = F$.

For completeness, if projections P, Q satisfy $\|P - Q\| < 1$, QP is bounded below on $P\mathcal{H}$ and maps onto $Q\mathcal{H}$; otherwise a nonzero $y \in Q\mathcal{H}$ orthogonal to its closed range would satisfy $Py = 0$, contradicting $\|(Q-P)y\| < \|y\|$. Thus $QP(PQP)^{-1/2}$ is a partial isometry from P to Q. Uniform continuity partitions $[0,1]$ into finitely many intervals with consecutive projections at distance less than one. Composing these partial isometries proves the asserted equivalence in the same algebra and commutant, without assuming finite rank.

For $z > 1 + a$, use the positive Schur complement of $z - JAJ$:

$$z - K - C^*(z - D)^{-1}C \geq \left(z - 1 - a - \frac{c_0^2}{z - 1 + b}\right) E.$$

Writing $z = 1 + a + s$ shows positivity when $s(b + a + s) > c_0^2$. The positive root is s_+, proving the upper endpoint by closedness of the spectrum. Its rationalized expression gives $s_+ \leq c_0^2/(b + a)$. The case of a zero block is included by the corresponding one-block inverse.

For the last assertion apply the resolvent Neumann series to the straight norm path from JAJ to A'. A self-adjoint perturbation of norm at most u has spectrum within distance u of the old spectrum. The two enclosing intervals remain separated because $2u < b - d_0$. The common-gap continuous calculus again provides a projection path and equivalence. This proof needs an actual operator-norm bound on the whole space, not a finite-section estimate. □

Corollary 5.2 (Conditional source deletion tolerances)**.** *For the source model compression, if $\|O\| \leq 1$, $OE_* = E_*$, the complementary upper bound is $1 - \chi$, and $\|(I - \Pi)E_*\|^2 \leq \beta$, then $2\beta < \chi$ preserves a gap before realization. Under an additional full regular-norm realization error η, it suffices for gap persistence that $2\beta + 2\eta < \chi$. For an actual copy satisfying (2) with $a = \alpha$, $b = \chi - \alpha$, and squared trimming leakage at most $\mathcal{S}$, the sufficient local condition is $2\alpha + 3\mathcal{S} < \chi$; a further self-adjoint norm error u is allowed if $2\alpha + 3\mathcal{S} + 2u < \chi$.*

Proof. Use the norm-one variant for the model, and the norm-two theorem for the actual copy. Substitution gives the stated inequalities. Equivalence and trace preservation only apply under the commutation and trace hypotheses specified below. These inequalities preserve a gap, not necessarily the source's narrower prescribed α-window. For that window one must also use (10). □

6 Equivariant Trace and a Quantitative Injection

Let an operator X commute with the left action on $\ell^2(\Gamma) \otimes \mathcal{K}$, where $\mathcal{K}$ has countable orthonormal basis (k_j). For $X \geq 0$ define

$$\mathrm{Tr}_\Gamma(X) = \sum_j \langle \delta_1 \otimes k_j, X(\delta_1 \otimes k_j)\rangle \in [0, \infty].$$

For an equivariant bounded map V between such multiples, $\mathrm{Tr}_\Gamma(V^*V) = \mathrm{Tr}_\Gamma(VV^*)$. To see this, expand the first side as the sum of squared matrix coefficients of V from identity inputs to all group outputs. Equivariance translates an output at g to the identity and the input to g^{-1}. Reindexing $g \mapsto g^{-1}$ gives the second side; Tonelli justifies the nonnegative countable sums. Projective phases, when the same multiplier is used on both sides, have absolute value one and disappear in this calculation. Polar maps remain equivariant: use the strong limit of $V(V^*V + n^{-1}I)^{-1/2}$ on its support. Consequently equivariant partial isometries preserve this trace, including infinite values.

This is not a statement about arbitrary Hilbert-space isometries. In particular, the source's ordered assigned domains need not be equivariant. Their orthogonal direct sum proves a norm and spectral statement; its dimension cannot simply be substituted for canonical trace. The source instead uses

$$\mathcal{L} = \bigoplus_Q \ell^2(V(Q)) \otimes \mathbb{C}^N, \quad \mathcal{E} = \bigoplus_Q E_Q, \quad \mathcal{A}\xi = (E_Q(\xi|_{V(Q)}))_Q.$$

The incidence identification is $\mathcal{L} \cong \ell^2(\Lambda) \otimes \ell^2(V(Y)) \otimes \mathbb{C}^N$; each pair consisting of a group vertex and a base type occurs once. This requires the actual indexed-copy construction, not an assumption that all copy stabilizers are trivial.

Writing e_v for the unnormalized diagonal trace at base type v, the weighted bound in the source is

$$\mathcal{S} = \sum_i 2^{C_\bullet H_i} \sum_{v \in A_i \cup B_i} e_v = o(1), \qquad \mathcal{S}_0 = \sum_v e_v \le \mathcal{S}.$$

Its finite-ball Cauchy–Schwarz argument yields $\sum_Q \left\| E_Q(\xi|_{I_Q}) \right\|^2 \le \mathcal{S} \|\xi\|^2$ for $I_Q = V(Q) \setminus D_Q$. Disjointness handles the retained restrictions, so $\|\mathcal{A}\| \le 1 + \sqrt{\mathcal{S}}$. The formula for $\mathcal{A}$, unlike that decomposition, is equivariant. With $\mathcal{K}_0 = \ker \mathcal{A}$, polar decomposition gives $\mathrm{Tr}_\Lambda(P_{\mathcal{K}_0^\perp}) \le \mathcal{S}_0$.

Theorem 6.1 (A physical frame bound for the trace injection)**.** *Let $T = T^*$ be bounded with $T \ge mI$. Let P reduce T and satisfy $TP \ge hP$. Suppose a closed subspace $\mathcal{K}_0$ satisfies $\langle y, Ty \rangle \le l \|y\|^2$ for $y \in \mathcal{K}_0$, where $m \le l < h$. Then*

$$\left\| P_{\mathcal{K}_0^\perp} x \right\|^2 \ge \frac{h - l}{h - m} \|x\|^2 \quad (x \in P\mathcal{H}). \tag{11}$$

The range is closed, and the inverse onto the range has norm at most $\sqrt{(h-m)/(h-l)}$. If both projections are equivariant, their polar map gives $\mathrm{Tr}_\Gamma(P) \le \mathrm{Tr}_\Gamma(P_{\mathcal{K}_0^\perp})$.

Proof. Since P reduces T, its two orthogonal components give $\langle y, Ty \rangle \ge h \|Py\|^2 + m \|(I-P)y\|^2$. For $y \in \mathcal{K}_0$, therefore, $\|Py\|^2 \le (l-m)\|y\|^2/(h-m)$. Taking adjoints shows the same bound for $P_{\mathcal{K}_0} P$. Pythagoras proves (11). A bounded-below operator has closed range and the stated inverse bound. Equivariance passes to polar decomposition, and trace monotonicity proves the last assertion. There is no finite-dimensional rank argument. □

For the source's final bounds, $h = 1 - 3\alpha$, $l = 1 - \chi + 3\alpha$, and $m = -2 - 3\alpha$ are valid because $\|T - C\| \le 3\alpha$ and $\|C\| \le 2$. Thus the source's injectivity step has the explicit stronger conclusion

$$\left\| P_{\mathcal{K}_0^\perp} x \right\|^2 \ge \frac{\chi - 6\alpha}{3} \|x\|^2 \quad (x \in p_0\mathcal{H}). \tag{12}$$

This is conditional on its spectral and all-copy form estimates. It does not manufacture an equivariant finite model. It supplies a physical lower frame margin in the sense of (3). The factor in (11) is sharp: in two dimensions take $T = \mathrm{diag}(h, m)$, P onto the first coordinate, and $\mathcal{K}_0$ spanned by $\sqrt{(l-m)/(h-m)}e_1 + \sqrt{(h-l)/(h-m)}e_2$. Its Rayleigh quotient is l and equality holds.

7 Scalar Compression and a Dimension-Free Inverse Cost

We rederive the source's scalar step rather than identifying a matrix corner with a scalar group algebra. Let $p \in M_N(C_r^*(\Lambda))$ be a projection with unnormalized trace $0 < a < 1/2$, put $K = \Lambda * \mathbb{F}_N$, and let $U = (\lambda(t_1), \dots, \lambda(t_N))$. Restriction to left Λ-orbits identifies the representation of $\mathbb{C}[\Lambda]$ on $\ell^2(K)$ with a direct sum of its regular representations. Thus $C_r^*(\Lambda) \subset C_r^*(K)$ is isometric and trace preserving. This fact, not an arbitrary representation, licenses regarding p as a matrix over $C_r^*(K)$.

Proposition 7.1 (Row margin and polynomial approximation)**.** *Set*

$$c_a = \sqrt{1-a} - \sqrt{a} > 0, \quad M_a = \min\{N, (1+\sqrt{a})^2\}, \quad A = pU^*Up.$$

Then $c_a^2 p \le A \le M_a p$. In its corner, A^{-1} exists and

$$V_s = UpA^{-1/2}, \qquad e = V_s V_s^* = (Up)A^{-1}(Up)^* \in C_r^*(K), \qquad \tau_K(e) = a. \tag{13}$$

Let $q = 1 - c_a^2/M_a \in [0,1)$ and, for an integer $n \geq 0$, put

$$S_n = M_a^{-1} \sum_{k=0}^{n} (p - A/M_a)^k, \qquad e_n = (Up) S_n (Up)^*. \tag{14}$$

Then

$$0 \leq e_n \leq e, \quad \|e - e_n\| \leq q^{n+1}, \quad 0 \leq \tau_K(e) - \tau_K(e_n) \leq a q^{n+1}, \quad \left\| e_n^2 - e_n \right\| \leq \min\{q^{n+1}, 1/4\}. \tag{15}$$

If $q^{n+1} < 1/2$, spectral rounding of e_n at $1/2$ equals e.

Proof. Use free-product normal words. Let $\mathcal{R}$ contain 1 and words whose first factor lies in $\mathbb{F}_N$, so each left Λ-orbit has the form Λv, $v \in \mathcal{R}$. For $\xi = p\xi$, define $m_v = \|\xi|_{\Lambda v}\|^2$ and $a_j = \tau(p_{jj})$. Restriction commutes with p, so the projection Cauchy–Schwarz bound is $|\xi_j(v)|^2 \leq a_j m_v$, with $\sum_j a_j = a$ and $\sum_v m_v = \|\xi\|^2$.

In coordinate j call a word bad if it starts with t_j^{-1}. Its initial letter uniquely identifies that coordinate and it belongs to $\mathcal{R}$. Hence $\|\xi_b\|^2 \leq a \|\xi\|^2$. Good outputs under U start with their respective t_j and are disjoint, so $\|U\xi_g\| = \|\xi_g\|$. For an output word h, weighted Cauchy–Schwarz gives

$$|(U\xi_b)(h)|^2 \leq a \sum_{j:\, t_j^{-1} h \text{ is bad}} m_{t_j^{-1} h}.$$

The map $(h, j) \mapsto t_j^{-1} h$ is injective on these bad inputs, because their first letter determines j. Summation gives $\|U\xi_b\| \leq \sqrt{a}\, \|\xi\|$. Consequently

$$c_a \|\xi\| \leq \|U\xi\| \leq (1 + \sqrt{a}) \|\xi\| .$$

The lower bound uses $\|\xi_g\| \geq \sqrt{1-a}\, \|\xi\|$; the upper uses $\|\xi_g\| \leq \|\xi\|$. Also $UU^* = N1$. Together these give the asserted two-sided bound on A. The improved upper bound retains the source's same good/bad decomposition, rather than discarding it in favor of N.

Functional calculus in the p corner gives $A^{-1/2}$ there. Thus $V_s^* V_s = p$, e is a scalar projection, and traciality on the finite row gives $\tau_K(V_s V_s^*) = \tau_{K,N}(V_s^* V_s) = a$. Writing $R_a = p - A/M_a$, we have $0 \leq R_a \leq qp$ and

$$A^{1/2} S_n A^{1/2} = p - R_a^{n+1}, \qquad e_n = V_s (p - R_a^{n+1}) V_s^* .$$

All claims in (15) follow from this exact identity, positivity, and $x(1-x) \leq 1/4$ on $[0,1]$. The spectrum of e_n is zero off $\operatorname{ran} e$ and contained in $[1 - q^{n+1}, 1]$ on that range, proving the rounding assertion. □

For any fixed margin $a \leq a_* < 1/2$, c_a is bounded below and $M_a \leq (1 + \sqrt{a_*})^2$ is bounded independently of N. The number of polynomial terms needed for $\|e - e_n\| \leq \zeta$, $0 < \zeta < 1$, can therefore be bounded independently of the matrix dimension. One sufficient condition is $n + 1 \geq M_a c_a^{-2} \log(1/\zeta)$, from $1 - x \leq e^{-x}$. This improves the source's displayed inverse series using the bound $A \leq Np$. It is an operator-polynomial degree bound, not a runtime or bit-complexity claim: p itself need not have finite group support. The cost still deteriorates as $a \uparrow 1/2$.

If Λ is finitely generated and torsion-free, then so is $\Lambda * \mathbb{F}_N$: a nontrivial finite-order element of a free product is conjugate into a factor, since positive powers of a cyclically reduced word of at least two alternating factors have strictly growing length. Thus (13) proves the desired scalar conclusion *conditional on the actual matrix projection and its unnormalized trace bound.* It does not construct that matrix projection.

Theorem 7.2 (Balanced scalar compression for every trace below one)**.** *Suppose $p \in M_N(C_r^*(\Lambda))$ is an actual projection with $0 < a = \tau_{\Lambda,N}(p) < 1$. Choose an integer*

$$M \geq N, \qquad M > \frac{a}{1-a}.$$

There is a scalar-unitary conjugate $\widetilde{p}$ of $p \oplus 0_{M-N}$ with $\tau(\widetilde{p}_{jj}) = a/M$ for every j. In $K = \Lambda * \mathbb{F}_M$, the row U_M of its M free generators is bounded below on $\widetilde{p}\ell^2(K)^M$ by

$$c_{a,M} = \sqrt{1 - a/M} - \sqrt{a} > 0. \tag{16}$$

It yields a scalar projection of trace a by the corner formula (13). Finite generation and torsion-freeness pass from Λ to K. The approximation estimates (14)–(15) hold with c_a replaced by $c_{a,M}$ and M_a replaced by $\min\{M, (1+\sqrt{a})^2\}$.

Proof. First retain the unequal diagonal traces in the proof of Proposition 7.1. A bad word v has one bad coordinate $j(v)$, not an arbitrary list. Hence

$$\|\xi_b\|^2 \leq \sum_{v \text{ bad}} a_{j(v)} m_v \leq a_{\max} \|\xi\|^2, \qquad a_{\max} = \max_j a_j.$$

The bad-output proof is unchanged and still gives $\|U\xi_b\| \leq \sqrt{a}\,\|\xi\|$. Therefore the lower margin is $\sqrt{1-a_{\max}} - \sqrt{a}$, positive if $a + a_{\max} < 1$. The upper row bound $1 + \sqrt{a}$ is unchanged.

Let $p' = p \oplus 0$ and form the ordinary $M \times M$ matrix $B = (\tau(p'_{ij}))$. It is positive: for $z \in \mathbb{C}^M$, $z^*Bz = \tau(\sum_{ij} \overline{z}_i p'_{ij} z_j) \geq 0$. Its ordinary trace is a. Choose a scalar unitary S with $SBS^* = \operatorname{diag}(\lambda_1, \ldots, \lambda_M)$, and let F be the normalized discrete Fourier matrix, so $|F_{ij}|^2 = 1/M$. For $U_0 = FS$, the projection $\widetilde{p} = U_0 p' U_0^*$ has diagonal traces $\sum_j |F_{ij}|^2 \lambda_j = a/M$. It still belongs to the same matrix reduced algebra and has the same unnormalized trace. The condition on M makes (16) positive. The normal-form proof, corner functional calculus, row trace identity, and polynomial error argument now apply verbatim with this positive margin. No scalar projection has been inferred from padding alone. □

This extends the source's *sufficient scalarization hypothesis* from $a < 1/2$ to $a < 1$ by an explicit construction. It does not improve the source's actual small-trace group construction or establish historical novelty: trace comparison in free products has a substantial literature [7, 8]. The scalar unitary uses exact trace data; its existence does not assert efficient access to that data. At $a = 1$, the faithful scalar trace would force the scalar projection to be the identity. For $a > 1$, a scalar projection of that trace is impossible. Thus the extended trace range has the correct natural boundary for a nontrivial scalar projection.

An exact trace-accounting example. Let $\Lambda = C_3 = \langle g \mid g^3 = 1 \rangle$ and

$$z = (1 + g + g^2)/3, \qquad p = 1 - z.$$

Convolution gives $z^* = z^2 = z$, hence p is a projection with canonical group trace $\tau(p) = 2/3$, its coefficient at the identity. This is not the normalized trace of an unrelated matrix example. Pad p to size three and conjugate by the normalized Fourier matrix. Its first column is flat, so every entry of $\widetilde{p}$ is $p/3$; in particular

$$\tau(\widetilde{p}_{jj}) = 2/9, \qquad \tau_{\Lambda,3}(\widetilde{p}) = 3(2/9) = 2/3, \qquad c_{2/3,3} = (\sqrt{7} - \sqrt{6})/3 > 0.$$

The theorem gives a scalar projection of trace $2/3$ over $C_3 * \mathbb{F}_3$, beyond the source's $a < 1/2$ sufficient range. The group has torsion; this is an illustration of the local transfer, not a counterexample to the torsion-free conjecture.

8 Perturbation Equivalence on the Full Gap Margin

Proposition 8.1 (Uniform contour homotopy)**.** *Let $B = B^*$ have a reducing projection F with $BF = F$ and complementary spectrum in $[-2, 1-b]$, where $0 < b < 1$. If $A = A^*$ and $u = \|A - B\| < b/2$, the upper-band projection of A is equivalent to F in $C^*(I, A, B, F)$. The equivalence preserves common equivariance and any associated identity trace. The upper spectrum lies in $[1-u, 1+u]$, while all other spectrum is at most $1 - b + u$.*

Proof. Put $r = b/2$ and $A_t = B + t(A - B)$ for $0 \le t \le 1$. The circle $|z - 1| = r$ has distance at least r from $\operatorname{spec} B$ and $\|(z - A_t)^{-1}\| \le (r - u)^{-1}$. Riesz integration gives projections F_t in the common algebra and

$$\|F_t - F_s\| \le \frac{ru}{(r-u)^2}|t - s|. \tag{17}$$

Indeed the resolvent identity supplies the integrand bound $u|t - s|/(r - u)^2$, and the circle length is $2\pi r$. Choose an integer $L > ru/(r - u)^2$. Consecutive projections at times j/L are at distance less than one. The polar maps proved in Theorem 5.1 compose to an equivalence from $F_0 = F$ to F_1. Spectral containment follows from the norm-perturbation resolvent argument. If $F = 0$, equivalence forces $F_1 = 0$; nonemptiness is not claimed in this case. □

With the maximal permitted radius $b/2$, the source's one-step close-projection estimate requires $u < b/4$ to prove equivalence. The homotopy reaches $u < b/2$, the band-separation threshold in this argument. It does not assert a stronger global estimate for $\|F_1 - F_0\|$. A common uniform choice of L also works for measurable direct-integral fields: resolvents, Riesz projections, corner inverse square roots, and the finite products are measurable. Trace equality is fiberwise only when all participating fields have the requisite common equivariance and trace representation.

9 Examples, Costs, and Unmet Obligations

9.1 A quadratic example without an off-diagonal block

Take $A = \operatorname{diag}(1, -2)$, E onto e_1, and J onto $v = \sqrt{1 - \varepsilon^2}e_1 + \varepsilon e_2$, $0 \le \varepsilon < 1$. Then $G = 1 - \varepsilon^2$, $Ve_1 = v$, and the compressed scalar is $1 - 3\varepsilon^2$. Its displacement is exactly quadratic; replacing it by the source's 20ε loses the decisive reducing-space structure. This example certifies the diagonal constant 3 under $\|A\| \le 2$, not sharpness of every simultaneous block bound.

9.2 What finite compression cannot establish

The actual Volume 21 proof [5] gives a nested graph-core example: $Ae_k = (k + 1/4)e_k$ on $\ell^2(\mathbb{Z})$ with domain $\sum_k (1 + k^2)|u_k|^2 < \infty$. The unit vector

$$\phi_m = \sqrt{\frac{m - 1/4}{2m}}e_m + \sqrt{\frac{m + 1/4}{2m}}e_{-m}$$

has Rayleigh quotient zero but full residual $\|A\phi_m\| = \sqrt{m^2 - 1/16}$. On the nested spaces spanned by e_k, $|k| \le n$, and ϕ_{n+1}, every compression has the spurious eigenvalue zero, while $\operatorname{dist}(0, \operatorname{spec} A) = 1/4$. Coordinate truncation proves graph-core density; it does not remove the omitted residual. This explains why neither finite spectral agreement nor strong resolvent convergence verifies M570's uniform regular-norm approximation.

9.3 Benefits and their prices

The exact Gram calculation separates quadratic upper-block leakage from linear coupling. The block norm estimate can be substituted into the source's contour proof; the Schur proof instead changes the sufficient gap criterion. Neither replacement automatically improves the eventual generating-set size, which depends on all preceding parameter choices. The all-copy result supplies an explicit stable injection rather than bare injectivity, conditional on the source form estimate and lower spectral bound.

The balanced scalar theorem broadens the admissible matrix trace range and gives a matrix-dimension-independent inverse degree for a fixed trace margin. Its price is padding, trace-based unitary mixing, and a possibly larger free factor. The polynomial acts on an actual reduced-algebra projection; evaluating or approximating that projection is a separate problem. No measured computational speedup is asserted.

9.4 Future extensions and obstacles

An anisotropic version can retain the full positive leakage $E-G$ and its interaction with A_E, rather than replace it by ε^2. It would require form inequalities in a common physical metric, not entrywise small matrix errors. A sharper all-copy comparison might retain local diagonal matrices instead of their scalar traces, but must still pay the overlap multiplicities and preserve equivariance. For approximate scalar inputs, one must first construct an actual projection in the correct reduced algebra with a rigorous trace interval and norm error. A floating nearly idempotent matrix does not meet those requirements.

The local methods do not establish torsion-freeness, uniform path recovery, the nilpotent fiber model, simultaneous voltage sampling, or the embedding and forest properties. Those remain source-specific dependencies of the asserted global theorem. Classical functional calculus, Schur complements, frame inequalities, and normal-form arguments are acknowledged as such; their use in Ram coordinates is not by itself a claim of novelty.

10 Evidence and Contribution Boundary

10.1 Current public context, checked October 8, 2026

The public repository README currently reports 719 manuscripts in 372 families, whereas this study preserves the original 722 manuscript identifiers. It explicitly describes mixed verification status [11]. Searches of the current formalization catalogue for "Kadison" and "285" returned no matching entry, and the attempted family-document URL returned 404. Those observations do not prove that no related formalization exists. We have not executed a Lean kernel check for M570 and make no machine-certification claim.

The current primary comparison literature counsels against a priority claim for scalarization. Dykema–Rørdam's first paper states its positive-K_0-cone conclusion for faithful tracial states under Avitzour and nuclearity hypotheses [7]; their second concerns infinitely many free factors with size conditions [8]. Neither abstract alone licenses dropping those hypotheses or replacing infinitely many factors by our specified finite row. Amrutam, Gao, Kunnawalkam Elayavalli, and Patchell prove strict comparison for reduced free-group algebras and, more generally, finitely generated acylindrically hyperbolic groups with trivial finite radical and rapid decay [9]. We do not establish all those assumptions for the source's group and do not import that theorem into our proof. The 2026 version of the work on selfless reduced free products and graph products provides further

relevant context [10]. The present claim is an explicit finite-row margin and its cost relative to `scalar:compression`, not a first existence result in free-product comparison theory.

10.2 Scope of the evidence

The report distinguishes literal source reading, analytic proof, finite exact checks, and numerical diagnostics. The accompanying M570 audit records the actual read scopes and file hashes. The local theorems are proved above independently of finite tests. Parent floating tests of rotated projections are useful crosschecks but are not substituted for these proofs or included in our own executed test count. No private corpus content is used in public search queries.

The realized contribution is the specific compression refinement, quadratic-gap consequence, quantitative equivariant injection, and balanced scalarization with inverse-cost bounds. Its internal score is C3/5, independent of the catalogue's applicability rating L4. This is neither external peer review nor global counterexample certification. Detailed scientific source dependencies and current public-status observations are recorded separately from this local mathematical conclusion.

References

[1] OpenAI, *A torsion-free counterexample to the Kadison–Kaplansky projection conjecture*, September 23, 2026, pinned repository manuscript, especially `scalar:compression`, `spec:compression`, `spec:perturbation`, and `spec:kernel-form`. https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-Torsion-Free-Counterexample-to-the-Kadison-Kaplansky-Projection-Conjecture-September-23-2026.

[2] M. Subbarao, private Ram foundation report, *Stable Adaptive Ram Frames*, actual source `RamFoundationTRs/TR-rev-1-old-tr248/TR-rev-1-old-tr248.tex`, `thm:parseval`, `eq:leak`, `thm:finite`, `eq:modelerror`, `thm:perturb`.

[3] M. Subbarao, private *Ram Calculus Books*, Volume 21, `chapters/ch01d-physical-metrics.tex`, physical Gram, metric symmetry, residual decomposition, and relative Gram bounds.

[4] M. Subbarao, private *Ram Calculus Books*, Volume 21, `chapters/ch01a-state-domain.tex`, bounded coordinate equivalence and transported physical inner products.

[5] M. Subbarao, private *Ram Calculus Books*, Volume 21, `chapters/ch01f-spectral-limits.tex`, graph-core pollution, strong versus norm resolvents, and domain-aware inversion.

[6] V. Kostrykin, K. A. Makarov, and A. K. Motovilov, *A generalization of the* $\tan 2\Theta$ *theorem*, 2003 preprint, https://arxiv.org/abs/math/0302020.

[7] K. J. Dykema and M. Rørdam, *Projections in free product* C^**-algebras*, Geom. Funct. Anal. 8 (1998), 1–16; erratum 10 (2000), 975. https://arxiv.org/abs/funct-an/9702016.

[8] K. J. Dykema and M. Rørdam, *Projections in free product* C^**-algebras, II*, Math. Z. 234 (2000), 103–113. https://arxiv.org/abs/math/9911009.

[9] T. Amrutam, D. Gao, S. Kunnawalkam Elayavalli, and G. Patchell, *Strict comparison in reduced group C^*-algebras*, Invent. Math. 242 (2025), 639–657, published September 18, 2025. https://link.springer.com/article/10.1007/s00222-025-01366-5.

[10] F. Flores, M. Klisse, M. Ó Cobhthaigh, and M. Pagliero, *Selfless reduced free products and graph products of C^*-algebras*, arXiv:2510.24675v3, August 5, 2026. https://arxiv.org/abs/2510.24675.

[11] OpenAI, public repository README and formalization catalogue, checked October 8, 2026. https://raw.githubusercontent.com/openai/math/main/README.md; https://raw.githubusercontent.com/openai/math/main/lean/formalization.yaml.